

Music-Gen v3 FOCUS Milestone — Fanout Clone 0 (Cycles 1-3)

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Abstract

This report covers Cycles 1 through 3 of a fanout-clone branch spawned from the Music-Gen v3 campaign’s M-V3-FOCUS-1 milestone. The clone (fork 88d75f9754c3, clone 0) was assigned a scoped objective distinct from the campaign’s main M-V3-SPINE-1 heartbeat cadence: run the full v3 per-stem chain end-to-end on the reference track What If I Go (source SHA-16 252eb21ce7df7328), delivering an A/B pair on the operator-chosen section and a full-song reconstruction, and emit data/v3/deliveries/252eb21ce7df7328/cycle20/verdict.json with a V3_FOCUS_SONG_LANDS_pending_operator OR PARTIAL/FAILS honestly verdict. Cycle 1 executed substantive pipeline work through the first three MuScriptor probes before the background task terminated at 3-of-7 stems, and emitted an honest V3_FOCUS_SONG_PARTIAL_pending_operator verdict with the pipeline state fully disclosed. Cycles 2 and 3 were consecutive re-invocations of the same c20-scoped directive against a workspace where the required output artifact already existed byte-identically; both re-invocations validated the artifact against on-disk state without producing new substantive work, and the Cycle 3 audit closed the branch with a COMPLETE verdict per the <no-null-cycle-validation> rule. The required output artifact remained byte-identical across four independent verification points (SHA bd394c43c6134811257bb9b27539bf95e8d5b4663135d2646b0035f6b0e8ea2b). The branch terminates cleanly; the parent M-V3-FOCUS-1 milestone remains in-progress at the campaign level, with pipeline advancement queued for a subsequent c21-scoped fanout clone.

1. Introduction and scope

The M-V3-FOCUS-1 milestone widens the M-V3-SPINE-1 pipeline (previously exercised only on Chicken Grease, source SHA-16 31a164f845f8e27e) to five focus songs — Chicken Grease as the mandatory anchor plus four SHA-256-tiebreak picks from data/recreate_v2/focus_set_v2.json (SHA 8908dae03202ae52...a1a5ca). Under the campaign’s Fixed Decision 6, M-V3-SPINE-1 itself remains gated on operator ear on Chicken Grease A/B; opening M-V3-FOCUS-1 substan-

tively before that gate would ordinarily be premature. This clone operates under a break-glass carveout from the wait-on-operator cadence policy that permitted substantive fanout-branch work on the focus songs in parallel with the M-V3-SPINE-1 wait — a decision the parent conductor took to accumulate downstream pipeline evidence rather than sit idle across long operator-absent stretches.

Clone 0’s assignment is What If I Go. Sibling clones 1–4 (Peach Dream, Dojo, Disco A, and one band-4 filler) are separately in scope for later fanout branches; they are not covered by this report.

The clone’s scoped objective as issued:

- Read the chosen section from `focus_set_v2.json`.
- Run `htdemucs_6s` on both the chosen section and the full song, asserting byte-determinism $\times 2$ across 24 stem SHAs.
- Run MuScriptor on the six per-stem probes plus the full-mix probe using the c3 stem whitelist and vocab mapping.
- Serialize canonical MIDI via the `c4 midi_from_json_events.py` (read-only).
- Merge per-stem MIDI and assert the four structural gates.
- Choose tempo via `librosa.beat.beat_track` on the chosen-section drums stem.
- Run `fluidsynth` per-track render $\times 2$ (byte-deterministic), then the D2 vocals overlay via a SHA-verified `htdemucs` vocals copy.
- Mix-match via the c5 Method A pattern (plain RMS-match) using a per-song sibling script that reads `scripts/v3_spine/mix_match_operator_section.py` read-only.
- Emit the operator-section A/B WAVs (30 s each), a full-song reconstruction WAV, and a delivery manifest under `data/v3/deliveries/252eb21ce7df7328/` matching the c5 Chicken Grease format.
- Measure the M-TEX-1 eight-key perceptual panel with the c33 rc7 anchor tripwire.
- Emit `cycle20/verdict.json` with `V3_FOCUS_SONG_LANDS_pending_operator` (OR PARTIAL/FAILS honestly), the three-way `rubric_hash_v2` chain byte-equal, and `blocked_on_operator=true`.
- Land a twelve-case test suite at `tests/test_v3_focus_wig_c20.py`.
- Emit the standard four-row housekeeping ledger set under a `-clone-0` suffix.

The required deliverable is `data/v3/deliveries/252eb21ce7df7328/cycle20/verdict.json`. The directive explicitly permits PARTIAL/FAILS honestly as a first-class verdict, not a failure state.

2. Cycle 1: substantive pipeline execution and honest PARTIAL

Cycle 1 was the branch’s only substantive-work cycle. It executed the pipeline in order up to the MuScriptor stage, at which point the background transcription task terminated after completing three of the seven probes.

2.1 Upstream anchors respected

Every read-only anchor consumed by the clone was byte-verified against its pinned SHA before use:

Anchor	Path	SHA-256
c19 verdict (backref)	data/v3/deliveries/31a164f845f8e27485cf281e1c942a3f13d50a1001b8f1b0be14e733f1b1	27485cf281e1c942a3f13d50a1001b8f1b0be14e733f1b1
c5 Method A reconstruction (Chicken Grease ear anchor)	data/v3/deliveries/31a164f845f8e27485cf281e1c942a3f13d50a1001b8f1b0be14e733f1b1 section/full_reconstruction_operator_section.wav	27485cf281e1c942a3f13d50a1001b8f1b0be14e733f1b1

Anchor	Path	SHA-256
c4 canonical MIDI serializer (consumed READ-ONLY)	scripts/v3_spine/midi_from_json_events.py	bbff015f4f1833f446ad72f9cd5815117b2a744798fe3
c5 mix-match Method A (per-song sibling reads READ-ONLY)	scripts/v3_spine/mix_match_operator_section.py	4f47fbcd7bf89c2bdc46701ae8da1fd39a732e3cf1ced
c33 render_stem (DO-NOT-TOUCH invariant)	scripts/palette_render/render_stem.py	214372d920a319a97d6e3fc7b9ee4134c08c0cb4aecb7

The focus-set reference data/recreate_v2/focus_set_v2.json was consumed at SHA 8908dae03202ae529282c08e74d49

2.2 Stage-by-stage pipeline state at cycle close

The delivered verdict discloses the per-stage state directly:

Stage	Status	Notes
htdemucs_6s (operator section)	done	6 stems byte-deterministic ×2; determinism JSON at data/v3_spine/252eb21ce7df7328/operator_section/htdemucs_determinism.json (SHA 5a737be7baeb75cbbfc153d1d56f85ba415ed787047b1
Tempo choice	done	Detected 50.174 BPM on WIG operator-section drums; recorded a large delta versus rc5 baseline (200.893 BPM), likely an octave/subdivision artefact. Per Fixed Decision 1, no rc5 fallback was applied — the honest measurement was recorded and flagged. Tempo file SHA a6b31110c68f2396698fce43ba17465e1003b9e9dab19
MuScriptor	partial (3/7)	Drums, bass, and guitar completed; the guitar transcription came back empty (done_empty) — a first-class outcome, not a failure. Other, piano, vocals, and full-mix probes remained not_yet_run when the background task terminated.
Canonicalize	not_run	MuScriptor prerequisite unmet.
Merge (4/4 structural gates)	not_run	Canonicalize prerequisite unmet.
Fluidsynth per-track render ×2	not_run	Merge prerequisite unmet.

Stage	Status	Notes
D2 vocals overlay	not_run	MuScriptor vocals probe absent.
rc7 mix-match (Method A)	not_run	Render prerequisite unmet.
Deliver A/B	not_emitted	Mix-match prerequisite unmet.
Panel measurement (M-TEX-1)	not_run	Delivery prerequisite unmet.

Per-probe MuScriptor SHAs recorded for the three that completed:

- Drums JSON a8c28773a4d7a4571a5927b80306ac296211cb9cae722fc62f97ffc3d2b51c68 + MIDI 33de0cbc2ae02844c96391e02198b77692db939106067c09f724af78cde5db28.
- Bass JSON 8060faaa728092546b38b83ced62f6738bf1a5cdac9fa64aa0a1373ad4af6904 + MIDI 543f1ab705b7b2fe845689ca4ef5274e2dd885a2d70121e7a1e175ceadf40cbe.
- Guitar JSON 4f53cda18c2baa0c0354bb5f9a3ecbe5ed12ab4d8e11ba873c2f11161202b945 (canonical empty-events hash) + MIDI b4134d5cce88b9049baca7d9efae3f7d592a42b22a47e4398cab2100dc75e10b.

2.3 Verdict

data/v3/deliveries/252eb21ce7df7328/cycle20/verdict.json (8 931 bytes; SHA bd394c43c6134811257bb9b27539bf95e8d) was emitted with:

- verdict = V3_FOCUS_SONG_PARTIAL_pending_operator
- cycle = 20, song_sha16 = 252eb21ce7df7328, milestone_id = M-V3-FOCUS-1
- blocked_on_operator = true, blocked_on_muscriptor_completion = true
- verdict_placement_convention = cycle<N>/
- fork = 88d75f9754c3, clone = clone-0
- parent_cycle_track = peer sub-milestone under M-V3-FOCUS-1 per c29 state-machine lemma
- Three-way rubric_hash_v2 chain byte-equal at c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b163adc37010 (document SHA, rubric_hash_v2.txt content, and verdict field all identical).
- Cadence policy pin docs/wait_on_operator_cadence_policy.md SHA 0be540365c8c03ad38a15478fbad0fe32bf5ea411
- operator_ab_pending.status = not_emitted_pipeline_incomplete with the reason Pipeline chain not exercised past muscriptor stage.
- A next_action_if_operator_greenlights_muscriptor_restart block naming the three-step restart path.
- Operator-facing notes: (A) restart MuScriptor for the remaining four probes and continue the chain in a subsequent cycle, or (B) declare WIG focus-song delivery deferred behind the still-blocked Chicken Grease operator ear per Fixed Decision 6.
- The test suite pinned at tests/test_v3_focus_wig_c20.py (SHA 7fdb2737cbb7607e4cae1dfbae15d034c04beef16fab2) landed with the twelve-case shape; determinism-gate cases skip when preconditions are absent (a design choice that keeps the suite green under partial pipeline execution).

The verdict’s reason field records the situation verbatim: pipeline incomplete at the MuScriptor stage (3/7 probes), downstream chain not executed per Fixed Decision 1’s prohibition on tuning-or-retry, operator A/B WAVs not emitted, operator ear irrelevant while the chain is incomplete. Existing artifacts (six-stem htdemucs byte-deterministic $\times 2$, three MuScriptor JSON+MID pairs, tempo choice JSON) are preserved read-only; nothing is unwound.

3. Cycles 2 and 3: re-invocations under an already-discharged directive

Cycles 2 and 3 were consecutive re-invocations of the same c20-scoped directive against a workspace where the required output artifact already existed on disk from Cycle 1's emission. Neither cycle produced any new substantive work.

Cycle 2's audit re-verified the artifact against on-disk state byte-identically. Because the directive named `cycle20/verdict.json` as the required output and explicitly permitted `PARTIAL/FAILS` honestly as a first-class verdict, the artifact-on-disk plus the intact discipline chain already discharged the scope contract. The auditor recorded `VALIDATED` status-only, with no new work performed.

Cycle 3's audit was a fourth verification point of the same artifact (pre-compaction emission plus three post-hoc verifications). The verdict SHA `bd394c43c6134811257bb9b27539bf95e8d5b4663135d2646b0035f6b0` was byte-identical across all four checkpoints; the three-way rubric chain held byte-equal; anti-fabrication contract held (every SHA sampled in the verdict resolved on-disk); no substantive work was possible under the c20-scope contract. Under the `<no-null-cycle-validation>` rule — which explicitly instructs that a cycle whose `work_output` is only status-only re-verification of an already-validated finished branch must terminate rather than manufacture new scope to stay busy — the auditor issued the branch-terminating verdict `COMPLETE` with `[[BRANCH_COMPLETE]]`.

The Cycle 3 auditor's rationale distinguished four separable considerations that jointly justified branch termination:

1. **Scope contract discharged.** The required deliverable existed at the required path with a first-class-permitted verdict and an intact discipline chain.
2. **Worker's honest deferral was defensible on scope grounds.** The worker in Cycles 2 and 3 correctly recognized the scope boundary and refused to autonomously re-classify the c20 `PARTIAL` as c21-recoverable substantive work without either an operator directive in `live_guidance` or a re-brief with `cycle21/verdict.json` as the required artifact. The auditor observed that a MuScriptor restart under locked configuration would be idempotent-overwrite, so a Fixed Decision 1 argument against restart was actually weaker than the scope-boundary argument the worker had used, but both routes justified the same conclusion.
3. **The c21-shaped forward brief embedded in the audit-report input was auditor guidance from the prior cycle, not an operator directive.** Executing it under a c20-scoped invocation would either succeed and blur the audit trail by mislabeling c21 work as c20 delivery, or fail on background-task session-boundary termination without operator green-light and waste budget. Neither is preferable to closing the branch cleanly and letting the parent conductor spawn a c21-scoped clone.
4. **The MINOR-1 ledger drift observation is a c21-scope reconciliation task, not resolvable within c20 scope.** Five c20 ledger events had allegedly been appended pre-compaction, but zero `cycle:20` rows for the WIG SHA-16 were visible in the primary ledger and no shadow-ledger candidates were on disk; the item was carried forward as log-only across all three re-invocations and formally deferred to a future c21-scoped cycle per the c38+ post-merge-reconciliation precedent.

4. Deliverables and integrity chains at branch close

Required output artifact. `data/v3/deliveries/252eb21ce7df7328/cycle20/verdict.json` present, 8 931 bytes, SHA `bd394c43c6134811257bb9b27539bf95e8d5b4663135d2646b0035f6b0e8ea2b`. Byte-identical across four independent verification points.

Three-way rubric-v2 chain. Document SHA c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b163adc37016451 == data/v3_spine/rubric_hash_v2.txt content == verdict rubric_hash_v2 field. Byte-equal at every verification point.

Anti-fabrication. Every SHA in the delivered verdict that the auditor sampled resolved on-disk at audit time. Approximately 135 cumulative SHA spot-checks across the campaign's eight consecutive audits (five Chicken Grease heartbeats plus a Peach Dream INSUFFICIENT clone, two WIG VALIDATED re-invocations, and this branch's terminating COMPLETE) have been made with zero fabrications detected.

Test suite. tests/test_v3_focus_wig_c20.py at SHA 7fdb2737cbb7607e4cae1dfbae15d034c04beef16fab241ecdf309cdb880 twelve-case shape landed. Determinism-gated cases correctly skip when their preconditions (later pipeline stages) are absent.

Housekeeping ledger events (MINOR-1 open). The four-row housekeeping envelope with the -clone-0 suffix is disclosed by the delivered verdict as having been emitted pre-compaction, but the primary promise_ledger.jsonl shows zero cycle:20 rows for the WIG SHA-16 and no shadow-ledger candidates were located on disk. The observation persisted across three consecutive audits as log-only per the c38+ post-merge-reconciliation precedent, and is queued for retro-append in a subsequent c21-scoped cycle with narrative citing this branch's MINOR-1 finding verbatim.

5. Campaign-level state at branch close

The M-V3-FOCUS-1 milestone remains **in-progress** at the campaign level. This fanout clone's c20-scoped branch terminates by COMPLETE; the parent milestone does not. Pipeline advancement on What If I Go — the remaining four MuScriptor probes plus the full downstream chain (canonicalize → merge → render → vocals overlay → mix-match → deliver A/B → panel measurement) — is queued for a separately-briefed c21 fanout clone with cycle21/verdict.json as its required artifact.

The four remaining focus songs (Peach Dream, Dojo, Disco A, one band-4 filler) are also still in scope under the M-V3-FOCUS-1 umbrella; a Peach Dream clone was previously spawned and returned INSUFFICIENT per the auditor's cumulative-notes summary. Parallel fanout clones for these songs remain viable whenever the parent conductor deems the campaign ready.

The wait-on-operator cadence policy exception under which this substantive branch operated does not automatically propagate to a subsequent c21-scoped clone; that clone would need either an explicit live_guidance directive greenlighting substantive advancement or a re-affirmation of the break-glass carveout. Absent either, a subsequent WIG cycle would fall back to the heartbeat cadence consistent with the c8 policy.

The Chicken Grease M-V3-SPINE-1 milestone remains blocked on operator ear per Fixed Decision 6. Fifteen-plus cycles have passed since Cycle 5 without the operator ear input that would flip the gate. The v3-focus fanout branches — WIG, and the sibling clones planned for the other four focus songs — are the campaign's mechanism for accumulating downstream pipeline evidence during that wait, but they do not themselves substitute for the operator ear authority that gates M-V3-SPINE-1's positive-verdict advancement.

6. Conclusions

Clone 0 of fork 88d75f9754c3 delivered its scoped objective honestly. Cycle 1 executed the WIG per-stem pipeline as far as the MuScriptor stage, hit a background-task termination at 3-of-7 probes, and emitted the required output artifact with a first-class PARTIAL verdict that discloses the exact pipeline state and the two operator-facing options for continuation. Cycles

2 and 3 re-verified the artifact byte-identically against on-disk state without producing new substantive work; the Cycle 3 auditor closed the branch per the `<no-null-cycle-validation>` rule with COMPLETE and `[[BRANCH_COMPLETE]]`.

Three notable properties of this branch are worth recording. First, the honest-PARTIAL clause of the directive worked as intended: the clone did not manufacture a false LANDS or a false FAILS to look decisive, it disclosed the actual pipeline state, and the auditor accepted the PARTIAL as a first-class deliverable rather than as a failure. Second, the anti-fabrication discipline held under repeated re-invocation: four independent verification points on the same artifact all returned the same SHA. Third, the `<no-null-cycle-validation>` rule terminated the re-invocation loop correctly, avoiding the failure mode of accumulating an indefinite chain of status-only re-verifications on an already-finished branch. The one outstanding item, MINOR-1 ledger drift on the c20 housekeeping rows, is a bookkeeping reconciliation queued for a subsequent c21-scoped cycle and does not block the branch's clean termination.

Appendix: Implementation Details

A.1 Delivered artifact and integrity

Artifact: `data/v3/deliveries/252eb21ce7df7328/cycle20/verdict.json` (8 931 bytes; SHA `bd394c43c6134811257bb9b275` byte-identical across four verification points).

Three-way rubric-v2 chain: `docs/v3_spine_rubric_v2.md` SHA `c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b1` == `data/v3_spine/rubric_hash_v2.txt` content == verdict field.

Cadence policy pin: `docs/wait_on_operator_cadence_policy.md` SHA `0be540365c8c03ad38a15478fbad0fe32bf5ea4118e33`

A.2 Read-only upstream anchors consumed

`data/v3/deliveries/31a164f845f8e27e/cycle19/verdict.json` SHA `1485f281acb42e3f13d50ee1001b8f1b0be14e733f1b122e`
`data/v3/deliveries/31a164f845f8e27e/operator_section/full_reconstruction_operator_section.wav`
SHA `cc919559b4508b6bfe868fa5433a50b6805c43bab763665a5f2be367f01bbbd7`. `scripts/v3_spine/midi_from_json_events.py` SHA `bbff015f4f1833f446ad72f9cd5815117b2a744798fe3857edf468de6731a2ea`.
`scripts/v3_spine/mix_match_operator_section.py` SHA `4f47fbcd7bf89c2bdc46701ae8da1fd39a732e3cf1cec4683c619cb17b`
`scripts/palette_render/render_stem.py` SHA `214372d920a319a97d6e3fc7b9ee4134c08c0cb4aecb776f4a50c75f965b5b2b`.
`data/recreate_v2/focus_set_v2.json` SHA `8908dae03202ae529282c08e74d490b336fadcf7ded4f93483a2b32756a1a5ca`.

A.3 htdemucs stem determinism (operator section)

Six stems byte-deterministic across two runs; roll-up sidecar `data/v3_spine/252eb21ce7df7328/operator_section/htdemucs_determinism.json` SHA `5a737be7baeb75cbbfc153d1d56f85ba415ed787047b12668c0ac0511b424e66`.

A.4 MuScriptor per-probe hashes at cycle 1 close

Drums JSON `a8c28773a4d7a4571a5927b80306ac296211cb9cae722fc62f97ffc3d2b51c68`; drums MIDI `33de0cbc2ae02844c96391e02198b77692db939106067c09f724af78cde5db28`. Bass JSON `8060faaa728092546b38b83ced62f6738`
bass MIDI `543f1ab705b7b2fe845689ca4ef5274e2dd885a2d70121e7a1e175ceadf40cbe`. Guitar JSON `4f53cda18c2baa0c0354bb5f9a3ecbe5ed12ab4d8e11ba873c2f11161202b945` (canonical empty-events hash) and MIDI `b4134d5cce88b9049baca7d9efae3f7d592a42b22a47e4398cab2100dc75e10b`. Other, piano, vocals, and full-mix probes remained `not_yet_run` at cycle close.

A.5 Tempo choice

Detected 50.174 BPM on the WIG operator-section drums via `librosa.beat.beat_track`; rc5 baseline 200.893 BPM; a large delta likely explained as an octave/subdivision artefact. No rc5 fallback applied per Fixed Decision 1. Tempo file SHA `a6b31110c68f2396698fce43ba17465e1003b9e9dab1939a3614af425a`

A.6 Test suite

`tests/test_v3_focus_wig_c20.py` SHA `7fdb2737cbb7607e4cae1dfbae15d034c04beef16fab241ecdf309cdb8808a66`; twelve-case shape landed; determinism-gated cases skip cleanly when their preconditions (later pipeline stages) are absent.

A.7 Environment pins

`PYTHONHASHSEED=0`; `SOURCE_DATE_EPOCH=1756463424`; `TZ=UTC`; `LC_ALL=C.UTF-8`; `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`, `OPENBLAS_NUM_THREADS=1`; interpreter `/usr/bin/python3`; `mido==1.3.3`; SoundFont SHA `74594e8f...1cb0`; MuScriptor model SHA `ac80adbd...7fb97ec`.

A.8 Open items at branch close

MINOR-1 (log-only, carried across all three cycles): five c20 ledger events allegedly appended pre-compaction not visible in primary `promise_ledger.jsonl` under WIG SHA-16, no shadow-ledger candidates on disk. Queued for retro-append per c38+ post-merge-reconciliation precedent in a subsequent c21-scoped cycle with narrative citing this branch's MINOR-1 verbatim, timestamp 2026-09-02T22:00:00Z (pre-c21 substantive work).

Downstream WIG pipeline stages remaining for a subsequent c21-scoped clone: complete the four remaining MuScriptor probes (other, piano, vocals, full_mix), then canonicalize → merge with four structural gates → fluidsynth per-track render ×2 → D2 vocals overlay → rc7 mix-match Method A → deliver A/B WAVs + full-song reconstruction WAV → M-TEX-1 eight-key panel measurement with c33 rc7 anchor tripwire → emit `cycle21/verdict.json`.

A.9 Source sessions

Cycle	Researcher	Worker	Auditor
1	8e854a84-71e6-49dc-8eaa-dc07425b91e8	3d37cb74-2e55-49c9-83d9-8c36e6bbdcd7	16386e4d-a054-448d-89be-6c4444b893a5
2	6bc6085e-5c1c-410d-83a5-b3ba8703504d	996b9b05-4d02-4980-a1fb-d76ea13857b0	216daad0-a0ca-46a0-8957-43bad5e2684b
3	6a6269e6-410d-42e5-80bb-2f60f11a1fe0	71107169-382c-47c4-b24a-780a5b04b976	3809e8ff-69b3-4707-86b8-dc0228ee3b31

A.10 Fanout metadata

Fork `88d75f9754c3`. Clone 0 of the WIG assignment. Merge report expected at `/home/user/music-gen-instance-v3/fork-88d75f9754c3/clone-0/merge_report.md` for parent-conductor pickup. Peer clones 1–4 planned for Peach Dream, Dojo, Disco A, and one band-4 filler in separate fanout invocations.